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UNIVERSITY OF GALWAY

Semester 2 Examinations 2023 / 2024

CT414 Marking Scheme

1. You are required to implement an Internet based College Course application system for the University using Java RMI. The server allows a client to login and retrieve an Application Form object that will then be completed on the client side. The client can then submit the completed Application Form object back to the server for validation and processing. The answer should include full source code for the following Java interfaces and classes:

a: Write a Java interface called *CourseServer* that provides remote methods for user login, based on a username and password, downloading an Application Form and submitting a completed Application Form. The login method returns a token, that is valid for some time period, that must be passed to the other two server methods as a session identifier. The methods to download and upload an Application Form object should use the interface type *ApplicationForm* as described in part b below. Include any required method exceptions. 4 MARKS

```
public interface CourseServer extends Remote {

        // Get sessionid from the server
        public int login(String username, String password) throws RemoteException,
                InvalidLogin;

        // Return an Application Form
        public ApplicationForm getForm(int token) throws RemoteException;

        // Submit a completed form
        public void submitForm(int ApplicationForm completed)
            throws RemoteException;
}
```

b: Write a Java interface called *ApplicationForm* that provides methods for the retrieval and answering of the kind of questions you would expect to find on an application form used to apply for a third level college course. At minimum it should include methods to allow the following information to be included in a completed application form: name, address, email and contact number of the applicant. It should also have a method to include a personal statement e.g. this could be used to include additional details about the applicant and a summary of their existing qualifications or results. Include any required method exceptions. 4 MARKS

```
public interface ApplicationForm extends Serializable {

        // Return general information about the form
        public String getInformation();

        // Set applicant details
        public void setApplicantDetails(String name, String address, String email,
                String contactNumber);

        // Set Personal Statement
        public void setPersonalStatement(String details);
```

```
        // Return all previously set details in name / value pairs
        public List<String, String> getContents();

        // Could aalso include accessor methods for individual pieces of information
}
```

c: Write the mainline Java server code, and the *CourseServer* implementation methods, required to initialise the server and register an instance of the *CourseServer* implementation class in the RMI Registry. A single valid username and password can be hardcoded into the server for the purposes of implementing the login method. You **do not** need to provide an implementation class for the *ApplicationForm* interface, you can assume that this has already been implemented in a class called *AppForm-v1*. The *CourseServer* implementation method to download an ApplicationForm therefore just needs to return an instance of the *AppForm-v1* class. The *CourseServer* implementation method to upload a completed should just save the completed form to a file. You **do not** need to provide implementations for any exception classes. 8 MARKS

Answer should include a main function that looks similar to the following:

```
public static void main(String[] args) {
        if (System.getSecurityManager() == null) {
                System.setSecurityManager(new SecurityManager());
        }
        try {
                String name = "CourseServer";
                CourseServer engine = new CourseEngine();
                CourseServer stub =
                   (ExamServer) UnicastRemoteObject.exportObject(engine, 0);
                Registry registry = LocateRegistry.getRegistry();
                registry.rebind(name, stub);
                System.out.println("CourseEngine bound");
        } catch (Exception e) {
                System.err.println("CourseEngine exception:");
                e.printStackTrace();
        }
}
```

d: Provide a simple command line client program that will interact with the server as follows: (i) Login to the server. (ii) Download an Application Form object. (iii) Call methods on the Application Form, as required to complete the form. (iv) Submit the completed Application Form back to the sever. 6 MARKS

Answer should include a main function that looks similar to the following:

```
public static void main(String args[]) {

        if (System.getSecurityManager() == null) {
                System.setSecurityManager(new SecurityManager());
```

```
}

CourseServer courseServer = null;
try {

        String name = "CourseServer";
        Registry registry = LocateRegistry.getRegistry(args[0]);
        courseServer = (CourseServer) registry.lookup(name);

} catch (Exception e) {

        System.err.println("Exception in ClientEngine");
        e.printStackTrace();

}

// Client code should then call some methods on the CourseServer…
```

e: What is the main advantage of defining *ApplicationForm* as an interface type and then using this in the remote methods? 3 MARKS

The main advantage is that new versions of the application form implementation class can be added in the future without having to recompile the client or main server code.

2.a: Write a short essay, approximately 300 words, on **one** of the following topics. The essay should include a full description of the topic and also discuss its advantages, disadvantages and competitor technologies (if applicable)

i. Enterprise Java Beans
ii. Cloud Computing
iii. Web Services 15 MARKS

Essay should take about 15 minutes and be about 300 words (about 1 written page).

b: Assume that you have been contracted by a large multinational company to develop an enterprise class client / server application that may be accessed by a large number of clients concurrently. You will therefore need to employ some form of load balancing in the design of the application. What type of load balancing systems would you recommend? In this context, describe both low-level and high-level load balancing mechanisms. Also provide some examples of real world systems or services that use the high-level load balancing mechanisms you have recommended. 10 MARKS

The answer should describe and evaluate possible information, placement and location policies used for load balancing. It should also reference a case study covered in detail in lectures where different load balancing approaches were tested and benchmarked. DNS rond robin and IP Anycast are suitable examples of high level load balancing mechanisms that were covered during the course.

3.a: What is unique about Node.js when compared to other server technologies like e.g. the Apache web server? How can a Node.js application efficiently handle reading and writing very large files e.g. where it is trying to make a copy of a file?
4 MARKS

Server side applications with a lot of I/O will benefit from using the event driven asynchronous way that Node.js runs e.g. Chat/Messaging, Real-time Applications, Intelligent Proxies, High Concurrency Web Applications. Thread based servers like Apache server will serve each web request as individual threads which doesn't fit well for concurrent connections. It should use streams to efficiently handle large data files.

b: Explain briefly the purpose of the **npm** utility. Describe the purpose and the effect of running the following command:

npm init 4 MARKS

npm is the package manager for Node.js platform. The command above will initialise and create the package.json file in the current directory. This file holds information about the version and module dependencies for a Node.js application.

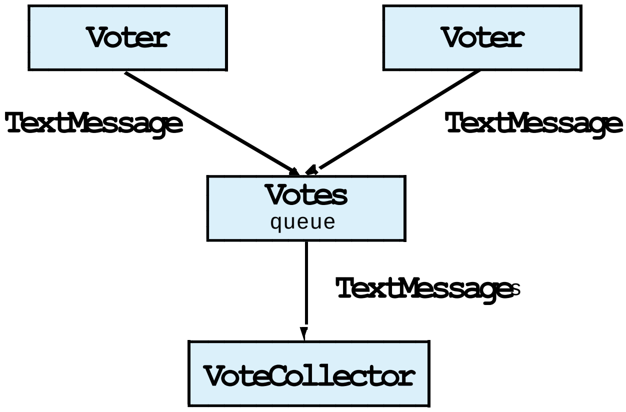
c: In the context of implementing a web server type application in Node.js what are the advantages of using Node.js with the Express framework? Write the Node.js code to implement a simple web server, using the Express framework, that responds with a simple text message when the URI /**main** is invoked.
7 MARKS

Express makes it easier to do things like routing web requests, delivery of static content like images and stylesheets, using "Middleware" e.g. request parsers, form processing, using simple forms of authentication, cookies and session manipulation.

```
// Simple web server using Express
var express = require('express');
var app = express();
app.get('/main, function (req, res) {
    res.send('Hello World!');
});
app.listen(3000, function () {
      console.log('Example app listening on port 3000!')
});
```

d: You are required to design an application that allows programmers to submit votes for their favourite programming language. Describe a suitable architecture and design for a distributed application that uses the Java Messaging Service (JMS) to submit these votes as messages to a queue. Another related application should similarly use JMS to consume these messages from the queue and tally votes. Full Java source code is not required but your answer should provide a full description of how the JMS could be used within the application. Also describe how the application might use the Java Naming and Directory Interface (JNDI) as part of this solution. 10 MARKS

This application should use the queue based point to point model with a related queue. A full answer should include and describe the following application components:



JNDI is used to locate the Votes queue and the related connection factory needed to connect to the queue. Answer should describe the usage of JNDI.

4 a: Explain the role of the Proxmox Virtualisation Environment. In this context, what is the difference between a Virtual Machine and a Container and which of these is faster to migrate to a different host on a Proxmox cluster? 6 MARKS

Proxmox Virtual Environment is an open-source server virtualization environment. It is a Debian-based Linux distribution with a modified Ubuntu LTS kernel and allows deployment and management of virtual machines and containers. A VM is a software-based environment geared to simulate a hardware-based environment, for the sake of the applications it will host. Containers also provide a way to isolate applications and provide a virtual platform for applications to run on but they do not provide a full VM. VMs are faster to migrate on Proxmox, as they do not need to be shutdown and restarted as part of the migration process.

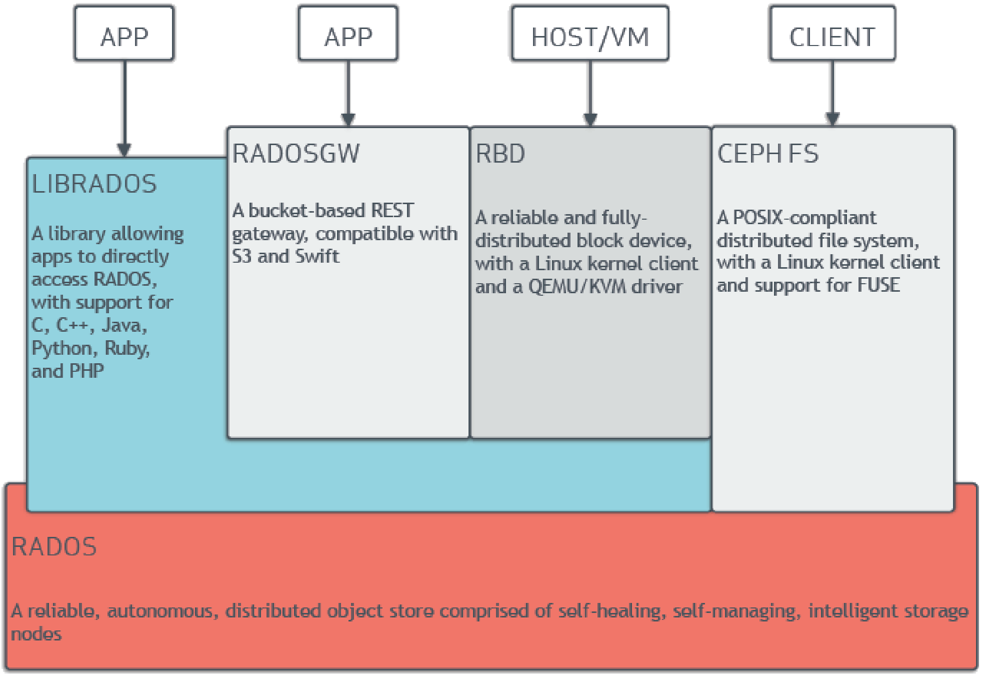
b: How is it possible to run Virtual Machines at near native speed using Kernel-based Virtual Machine (KVM) infrastructure? 4 MARKS

Answer should mention that KVM uses CPU extensions for virtualization via a linux kernel module that make it possible for VMs to run at near native speed.

c: What is the purpose of the Ceph storage platform and what advantages does it have over traditional RAID based storage? Describe the high level architecture and the main components of the Ceph storage platform. Include in this description details about the following items: Ceph Network, Object Storage Devices and Ceph Pools. 8 MARKS

Ceph is open source which means that no licensing fees are involved for using Ceph. As a result, Ceph will always be cheaper than proprietary storage options. Ceph storage is software-defined and is not dependent on expensive hardware. With Ceph, commodity hardware can be used to build a storage system that is highly scalable.

Ceph can be installed on common server hardware, instead of top-grade specialized storage hardware. Answer could include a diagram like the following:



d: What are the advantages of grouping physical servers into a cluster? How does a Proxmox cluster implement High Availability and what might cause a Virtual Machine migration to fail? 7 MARKS

A Proxmox High Availability Cluster enables the definition of high availability for virtual machines. In simple terms, if a virtual machine (VM) is configured as HA and the physical host fails, the VM is automatically restarted on one of the remaining Proxmox VE Cluster nodes. The Proxmox cluster technology is based on proven Linux HA technologies, providing stable and reliable HA service. A VM migration could fail if the same resources or processor settings are not available on the target host.

5.a: What types of services are typically available from commercial Cloud Computing providers? Provide some examples of each of these services in your answer.
5 MARKS

The answer should include a description and examples of Software as a Service (SAAS), Platform as a Service (PAAS) and Infrastructure as a Service (IAAS).

b: Suppose you work for a social media company that collects a lot of very large data sets e.g. web logs or other application related data that needs to be stored and analysed. Also assume that the company has access to large scale computing resources based in multiple data centres. Explain how using the Apache Hadoop Distributed File System (HDFS) and its related facilities might help in solving the storage and analysis requirements of the company. Your answer should include a high-level architectural diagram outlining how the HDFS is typically organised in terms of data block storage. Discuss the specific advantages of this approach over using traditional database systems for this type of data. 10 MARKS

Hadoop provides a reliable shared storage (HDFS) and analysis system (MapReduce). Hadoop is highly scalable and unlike the relational databases, Hadoop scales linearly. Due to linear scale, a Hadoop Cluster can contain tens, hundreds, or even thousands of servers. he Hadoop Distributed File System (HDFS) is the primary data storage system used by Hadoop applications. It employs a NameNode and DataNode architecture to implement a distributed file system that provides high-performance access to data across highly scalable Hadoop clusters. While storing the data onto HDFS, it is stored as large blocks of a specified size (default is 128MB) and it may be replicated.

c: Describe in detail the MapReduce programming model. Outline the architecture for a MapReduce application that could be used to index a large number of text files by the individual words present in each file. Full source code for the application is not required but your answer should include the data structures that could be used and also clearly explain the purpose and functionality of the map() and reduce() functions in solving this problem. What kind of threading model makes most sense in terms of running the map() and reduce() functions?
10 MARKS

MapReduce is a processing technique and a program model for distributed computing based on java. The MapReduce algorithm contains two important tasks, namely Map and Reduce. Map takes a set of data and converts it into another set of data, where individual elements are broken down into tuples (key/value pairs). Secondly, the reduce task takes the output from a map as an input and combines those data tuples into a smaller set of tuples. As the sequence of the name MapReduce implies, the reduce task is always performed after the map job. The major advantage of MapReduce is that it is easy to scale data processing over multiple computing nodes. Answer should also include the data structures that could be used as part of solution to the problem described above. Normally both the map() and reduce() operations can be run in parallel on separate threads.